

Sensing Gravitational Gradient via a Magnonic System

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Measuring gravitational gradient at the attohertz level pushes the boundaries of quantum metrology. We propose a rotating magnonic sensor operating at a second-order exceptional point (EP) to achieve this precision. By exploiting the square-root topological singularity of a non-Hermitian YIG-cavity system, we demonstrate an intrinsic signal amplification of $\eta \sim 10^{11}$, converting the relativistic 10^{-17} Hz shift into a microhertz spectral splitting. We present a rigorous Langevin noise analysis, showing that despite the Petermann factor enhancement of quantum noise ($K \sim 10^{11}$), the high photon flux and mechanical up-conversion strategy yield a signal-to-noise ratio (SNR) near to $\text{SNR} \approx 1$ for integration times $\tau = 1000$ s.

I. INTRODUCTION

Gravity gradient sensing provides numerous advantages and applications. It excels at locating mineral deposits [1], oil, and gas [2] by detecting subtle variations in subsurface density [3]. In addition, this technology is invaluable for creating detailed geological maps, revealing hidden underground features and fault lines [4]. It enhances the accuracy of navigation systems in submarines and spacecraft [5]. Furthermore, gravity gradient sensing plays a crucial role in civil engineering, as it helps assess ground stability for infrastructure projects such as tunnels and bridges [6].

A classic gravity gradiometer, such as the torsion balance type, can achieve a sensitivity of approximately 1 Eötvös but requires extensive measurement time at a single location [7]. Modern advancements aimed to improve speed and efficiency [8]. For instance, the rotating gravity gradiometer invented by Forward in the 1970s, used rotating proof masses to enhance sensitivity and was intended for moving-base applications [9]. Another example is the floated gravity gradiometer, which utilizes a neutral buoyancy system to reduce noise and improve accuracy, making it suitable for use in satellites and aircraft [10]. The full-tensor gravity gradiometer, developed in the 1980s by Bell Aerospace, utilized multiple accelerometers to measure the entire gravity gradient tensor, with applications in submarine navigation and airborne gravity surveys [11] [12]. Forward also conducted early experimental investigations into gravitational shielding and proposed that cryogenic cooling techniques could reduce instrumental noise and thereby enhance the sensitivity of

gravity gradiometers [8]. A notable design by Chobotov in 1968 involved a radially vibrating, rotating gravitational gradient sensor [13]. Additionally, the Falcon system, developed in collaboration with Lockheed Martin and BHP Billiton, performed the first commercial airborne gravity gradient survey in 1999 [14].

The first quantum gravity gradiometer, developed in Kasevich's laboratory, utilized laser-manipulated atom interferometry [15]. This innovative device employed a light-pulsed magneto-optical atom trapping technique to measure the absolute accelerations of two cesium atom ensembles, each free-falling along a ballistic trajectory. By dividing the difference in these measurements by the baseline distance between the ensembles, the gravity gradient was determined. Therefore, eliminating sensor misalignment errors and many systematic errors inherent to traditional gradiometers. The cesium atoms' uniform properties made the gradiometer immune to temperature and magnetic field variations. Calibration relied on the wavelengths of frequency-stabilized laser beams, ensuring identical references for both atomic proof masses. Subsequent advancements, including a patent filed by Lockheed Martin Corporation [16] and further research by Carraz et al. [17] and Sorrentino et al. [18], have improved sensitivity and expanded the application potential of these quantum gravity gradiometers.

When a network of coupled spins is excited, it generates a spin wave known as a magnon. These quanta have significant applications, including quantum sensing. Specifically, Magnons are utilized in detecting axions [19], where recent advancements in Ferromagnetic haloscopes explore their interaction with electrons via spin excitations in magnetic materials, enhancing sensitivity and bandwidth across various axion mass ranges. This experimental setup operates cryogenically, setting new limits on axion-electron coupling strength and discussing future improvements aligned with axion models. In another application, Magnons contribute to detecting gravitational waves [20], targeting frequencies above 10 kHz. Proposed detectors utilize magnonic systems to

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reinterpret data from axion experiments, enhancing sensitivity to GHz gravitational waves through innovative detector designs and theoretical frameworks. .

In this work, we address this question by considering a single cavity-based configuration in which a yttrium iron garnet (YIG) ellipsoid is strongly coupled to a microwave cavity mode, forming a hybrid cavity–magnon system. We show that a weak gravitational-gradient perturbation can be mapped onto a shift of the magnon resonance frequency, providing a direct spectroscopic readout channel. To enhance the response to such an extremely small perturbation, we further engineer the hybrid system in a gain-assisted non-Hermitian regime and tune it close to a second-order exceptional point. In this regime, the gravitationally induced magnon-frequency shift is converted into an enhanced hybrid-mode splitting, which serves as the measurable signal. To quantify the detector performance, we model the open-system dynamics using quantum Langevin equations together with the input–output formalism. This allows us to compute the linear response, the measurable output spectrum, and the corresponding noise floor. We then evaluate the impact of the Petermann-factor noise enhancement near the exceptional point and estimate the achievable signal-to-noise ratio of the proposed gradiometric scheme.

In the following sections, we first review the basic theory of magnon–gravity coupling and then introduce the proposed experimental setups. Next, we analyze the magnon frequency shift and the gain-assisted non-Hermitian dynamics of the system. Finally, we study the amplification of gravitational perturbations at the exceptional point (EP), followed by an evaluation of quantum noise and the overall sensitivity.

II. REVIEW OF MAGNON–GRAVITY COUPLING

To understand the working principles of a gravitational gradient detector based on magnonic systems, we begin by considering a ferromagnetic sample in an external magnetic field. Its microscopic spin dynamics can be described by the Heisenberg model (in natural units) [21]

$$H_{\text{spin}} = -2\mu_B B_z \sum_i \hat{S}_{(i)}^z - \sum_{i,j} J_{ij} \hat{S}_{(i)} \cdot \hat{S}_{(j)}, \quad (1)$$

where the Bohr magneton $\mu_B = e\hbar/2m_e$ (or $\mu_B = e/2m_e$ in natural units) is defined by the electron charge e and mass m_e , J_{ij} denotes the exchange interaction strength between spins, $\hat{S}_{(i)}$ is the spin operator at site i , $\hat{S}_{(i)}^z$ is its z -component, and B_z is the applied static magnetic field.

When the ferromagnetic sample is placed in a gravitational field, the local curvature of spacetime slightly perturbs the magnetic interaction energy of the spins. Following the treatment introduced in Ref. [20], the corresponding effective Hamiltonian for the magnon–gravity

interaction can be written as

$$H_{\text{gm}} = -\mu_B B_a \hat{S}_{(i)}^b Q_{ab}, \quad (2)$$

where B_a is the component of the applied magnetic field along direction a , and Q_{ab} is a tensor characterizing how gravity modifies the local magnetic environment within the sample. The explicit form of Q_{ij} is given by

$$Q_{ij} = -\frac{2\kappa}{3} \delta_{ij} \ddot{h}_{kl} \Big|_{x=0} x^k x^l - \frac{1}{6} (h_{il,kj} + h_{kj,il} - h_{kl,ij} - h_{ij,kl}) \Big|_{x=0} x^k x^l, \quad (3)$$

where $\kappa = \sqrt{16\pi G/c^4}$ (or $\kappa = \sqrt{16\pi G}$ in natural units), and the coordinates x^k and x^l represent spatial positions inside the YIG ellipsoid. The quantity $\kappa h_{ij} = g_{ij} - \eta_{ij}$ defines h_{ij} as the spatial part of the metric perturbation in the linearized gravity approximation, with g_{ij} the full spatial metric and η_{ij} the flat Minkowski background [20]. This general expression for h_{ij} will later be applied to the case of Earth’s gravitational field to analyze how the local spacetime geometry affects the magnonic modes of the YIG ellipsoid. The factors $x^k x^l$ in Eq. (3) encode the position dependence of the gravitational influence across the sample, which is essential for understanding how gravity couples to magnonic excitations. The second spatial derivative of the metric perturbation is defined as

$$h_{il,kj} = \frac{\partial^2 h_{il}}{\partial x^k \partial x^j}, \quad (4)$$

and $\ddot{h}_{kl}$ denotes the second time derivative of h_{kl} . Therefore, the interaction Hamiltonian describing the coupling between the gravity gradient and the spin system is given by

$$\begin{aligned} H_{\text{int}} &= H_{\text{gm}} + H_{\text{spin}} \\ &= -\mu_B B_z \sum_a (2\delta_{za} + Q_{za}) \sum_i \hat{S}_{(i)}^a \\ &\quad - \sum_{i,j} J_{ij} \hat{S}_{(i)} \cdot \hat{S}_{(j)}. \end{aligned} \quad (5)$$

The Hamiltonian (5) can be rewritten by using the Holstein-Primakoff transformation [22]

$$\hat{S}_{(i)}^z = \frac{1}{2} - \hat{B}_i^\dagger \hat{B}_i, \quad (6a)$$

$$\hat{S}_{(i)}^+ = \sqrt{1 - \hat{B}_i^\dagger \hat{B}_i} \hat{B}_i, \quad (6b)$$

$$\hat{S}_{(i)}^- = \hat{B}_i^\dagger \sqrt{1 - \hat{B}_i^\dagger \hat{B}_i}. \quad (6c)$$

where bosonic operators $\hat{B}_i$ and $\hat{B}_i^\dagger$ satisfy commutation relations $[\hat{B}_i, \hat{B}_j^\dagger] = \delta_{ij}$ and $\hat{S}_{(j)}^\pm = \hat{S}_{(j)}^x \pm i\hat{S}_{(j)}^y$ are the ladder operators. It is straightforward to check that the SU(2) algebra, $[\hat{S}^i, \hat{S}^j] = i\epsilon_{ijk} \hat{S}^k$ ($i, j, k = x, y, z$), is satisfied even after the transformation (6). We note that

$\hat{B}_i^\dagger \hat{B}_i$ represents the particle numbers of the boson created by the creation operator $\hat{B}_i^\dagger$. The bosonic operators describe spin waves with dispersion relations determined by B_z and J_{ij} . Furthermore, provided that contributions from the surface of the sample are negligible, one can expand the bosonic operators by plane waves as

$$\hat{B}_i = \sum_k \frac{e^{-i\mathbf{k}\cdot\mathbf{r}_i}}{\sqrt{N}} \hat{b}_k, \quad (7)$$

where $\mathbf{r}_i$ is the position vector of the i th spin and N denotes the total number of spins (lattice sites) in the sample. The excitation of the spin waves created by $\hat{b}_k^\dagger$ is called a magnon. We now rewrite the Hamiltonian (5) with the Holstein-Primakoff transformation (6) and then we only focus on the homogeneous mode of magnons, i.e., $\mathbf{k} = 0$ mode. Then, the second term in the total Hamiltonian (5) is irrelevant because it does not contribute to the homogeneous mode. Furthermore, because Q_{zz} does not contribute to the resonance of the spins, namely excitation of magnons, we will drop it. Finally we have

$$H_{\text{gm}} = \mu_B B_z \sum_i \left[(2 + Q_{zz}) \left(\frac{1}{2} - \hat{B}_i^\dagger \hat{B}_i \right) + \frac{\hat{B}_i + \hat{B}_i^\dagger}{2} Q_{zx} + \frac{\hat{B}_i - \hat{B}_i^\dagger}{2i} Q_{zy} \right]. \quad (8)$$

In order to measure Earth's gravity and gravity gradient, we replace h_{ij} in equation (3) with h_{ij} corresponding to the Earth's gravitational field [23]

$$h_{ij} = \begin{pmatrix} 2\Phi & 0 & 0 \\ 0 & 2\Phi & 0 \\ 0 & 0 & 2\Phi \end{pmatrix}, \quad (9)$$

where Φ is the gravitational potential of (spherically symmetric) Earth

$$\Phi = -\frac{GM}{|\mathbf{R} + \mathbf{x}|}, \quad (10)$$

where G is the gravitational constant, R is the radius of the Earth, and $\mathbf{x}$ is the distance of the center of the ellipsoid from the surface of the Earth. We substitute h_{ij} from Eq. (9) into Eq. (3) and obtain Q_{ij} for our problem. Therefore, the tensors Q_{zx} , Q_{zy} , and Q_{zz} are given as

$$Q_{zx}(x, y, z) = -\frac{1}{3} \left(\Phi_{,xx} xz + \Phi_{,yx} yz + \Phi_{,zy} xy + \Phi_{,zz} xz - \Phi_{,zx} y^2 \right), \quad (11)$$

and the tensor Q_{zy} is

$$Q_{zy} = -\frac{1}{3} \left(\Phi_{,xy} xz + \Phi_{,yy} yz + \Phi_{,zx} yx + \Phi_{,zy} y^2 + \Phi_{,zz} yz - \Phi_{,xy} x^2 - \Phi_{,yy} y^2 \right), \quad (12)$$

and finally the tensor Q_{zz} is

$$Q_{zz} = -\frac{1}{3} \left(\Phi_{,xx} x^2 + 2\Phi_{,xy} xy + \Phi_{,yy} y^2 - \Phi_{,zz} (x^2 + y^2) \right). \quad (13)$$

Substituting Eqs. (11), (12), and (7) into Eq. (8), we obtain the magnon-gravity coupling Hamiltonian expressed in terms of the bosonic operators. To properly account for the spatial variation of the fields over the YIG sample, the discrete sum over lattice sites is replaced by a continuum integral [20]

$$\sum_i f(\mathbf{r}_i) \simeq \frac{1}{L^3} \int_{V_{\text{YIG}}} f(\mathbf{r}) d^3r, \quad (14)$$

where L denotes the lattice constant. Consequently, the total number of spins in the sample, N , is given by

$$N = \frac{1}{L^3} \int_{V_{\text{YIG}}} d^3r = \frac{4\pi abc}{3L^3}, \quad (15)$$

in which a, b , and c represent the semi-axes of the YIG ellipsoid along the x, y , and z directions, respectively. Due to the symmetry of the ellipsoid (centered at the origin and aligned with its principal axes), all mixed spatial integrals vanish, i.e., $\int xz d^3r = \int yz d^3r = \int xy d^3r = 0$. Using this continuum approach, we define the volume-averaged components of the gravity gradient tensor, denoted by $\langle Q_{ij} \rangle$, as

$$\langle Q_{ij} \rangle = \frac{1}{V_{\text{YIG}}} \int_{V_{\text{YIG}}} Q_{ij}(\mathbf{r}) d^3r. \quad (16)$$

These averaged quantities represent the effective gravitational field experienced by the magnon mode across the entire YIG crystal.

Since constant terms only shift the zero-point energy and do not affect the system dynamics, they can be omitted. By substituting Eq. (7) into Eq. (8) and neglecting the constant contributions, the Hamiltonian for the magnon-gravity interaction becomes

$$H_{\text{gm}} = \mu_B B_z \left[\left(2 + \langle Q_{zz} \rangle \right) \left(\frac{N}{2} - \hat{b}^\dagger \hat{b} \right) + \frac{\sqrt{N}}{2} \langle Q_{zx} \rangle (\hat{b} + \hat{b}^\dagger) + \frac{\sqrt{N}}{2i} \langle Q_{zy} \rangle (\hat{b} - \hat{b}^\dagger) \right]. \quad (17)$$

Finally, the interaction Hamiltonian in SI units can be expressed as

$$H_{\text{gm}} = \mu_B B_z \left[\left(2 - \frac{gl_z^2}{35Rc^2} \right) \left(\frac{N}{2} - \hat{b}^\dagger \hat{b} \right) - \frac{\sqrt{N}}{2} \frac{gl_x^2}{35Rc^2} (\hat{b} + \hat{b}^\dagger) + \frac{\sqrt{N}}{2i} \frac{gl_y^2}{35Rc^2} (\hat{b} - \hat{b}^\dagger) \right]. \quad (18)$$

In this expression, l_x, l_y , and l_z are effective geometric factors that depend on the ellipsoid dimensions. The explicit derivations of these parameters in terms of the semi-axes a, b , and c are provided in Appendix A.

III. OUR SETUP

A schematic of the proposed system is shown in Fig. 1. The setup consists of a microwave cavity containing a small YIG sample subjected to a uniform bias magnetic field along the z direction. The collective spin excitations of the YIG crystal form a magnon mode that coherently interacts with the cavity photon mode. In the presence of a gravitational gradient, the magnon mode acquires an additional coupling that enables the detection of gravitational effects through the cavity output field.

A. Dynamics of the System

For this system, the total Hamiltonian can be written as [19]

$$H_{\text{tot}} = H_a + H_m + H_{am} + H_{gm}, \quad (19)$$

where H_a describes the energy of the cavity photons, H_m corresponds to the magnon mode of the YIG sample, and H_{am} represents the interaction between photons and magnons. The Hamiltonian of the cavity field is

$$H_a = \omega_a \hat{a}^\dagger \hat{a}, \quad (20)$$

where $\hat{a}$ ($\hat{a}^\dagger$) is the annihilation (creation) operator of the cavity photon mode. The Hamiltonian of the magnon mode in the YIG crystal is

$$H_m = \omega_m \hat{b}^\dagger \hat{b}, \quad (21)$$

where $\hat{b}$ ($\hat{b}^\dagger$) denotes the annihilation (creation) operator of magnons. The photon-magnon interaction is described by

$$H_{am} = g_{am}(\hat{a}^\dagger + \hat{a})(\hat{b} + \hat{b}^\dagger), \quad (22)$$

where g_{am} denotes the magnon-cavity coupling strength.

In a realistic experiment, both the cavity and magnon modes experience dissipation and therefore must be treated as an open quantum system. The cavity field is coupled to an external port used for probing the system, while both the cavity and magnon modes interact with their respective thermal environments. The dynamics of such open systems can be described using the Heisenberg-Langevin formalism. The general form of the Langevin equation for an arbitrary system operator f is given by [24]

$$\begin{aligned} \frac{df}{dt} = & -i[f, H] + \sum_n \left[\left(\frac{\kappa_n}{2} g^\dagger + i\sqrt{\kappa_n} a_n^{\text{in}\dagger} \right) [f, g] \right. \\ & \left. - [f, g^\dagger] \left(\frac{\kappa_n}{2} g + i\sqrt{\kappa_n} a_n^{\text{in}} \right) \right], \end{aligned} \quad (23)$$

where f and g are arbitrary system operators, κ_n is the dissipation rate associated with the n -th bath, and a_n^{in} represents the corresponding input noise field. On the

other hand, the total Hamiltonian of our magnon cavity system can be written as [25]

$$H_{\text{tot}} = \omega_a \hat{a}^\dagger \hat{a} + \omega_m \hat{b}^\dagger \hat{b} + g_{am}(\hat{a}^\dagger + \hat{a})(\hat{b} + \hat{b}^\dagger) + H_{gm}, \quad (24)$$

where H_{gm} is the magnon-gravity interaction Hamiltonian given in Eq. (18). By substituting the total Hamiltonian H_{tot} into the general Langevin equation (23) and choosing $f = g = \hat{a}$ as the cavity photon annihilation operator, we obtain

$$\frac{d\hat{a}}{dt} = -i\omega_a \hat{a} - ig_{am} \hat{b} - \frac{\kappa_a}{2} \hat{a} + \sqrt{\kappa_a} \hat{a}_{\text{in}}, \quad (25)$$

where κ_a is the damping rate of the cavity photons.

Similarly, applying Eq. (23) to the magnon operator $\hat{b}$ yields

$$\begin{aligned} \frac{d\hat{b}}{dt} = & -i \left[\omega_m - \mu_B B_z \left(2 - \frac{gl_z^2}{35Rc^2} \right) \right] \hat{b} - ig_{am} \hat{a} \\ & - \frac{\sqrt{N}}{2} \mu_B B_z \left(\frac{gl_x^2}{35Rc^2} (\hat{b} + \hat{b}^\dagger) + \frac{1}{i} \frac{gl_y^2}{35Rc^2} (\hat{b} - \hat{b}^\dagger) \right) \\ & - \frac{\kappa_m}{2} \hat{b} + \sqrt{\kappa_m} \hat{b}_{\text{in}}, \end{aligned} \quad (26)$$

where $\hat{b}_{\text{in}}$ is the magnon input noise field and κ_m denotes the magnon damping rate in the YIG sample.

B. Experimental Setup

The proposed sensing element consists of a YIG crystal coupled to a two-post re-entrant microwave cavity. The cavity is operated in a cryogenic environment and biased by a static magnetic field so that the uniform magnon mode can be tuned close to the microwave cavity mode. The YIG sample is placed near the magnetic-field antinode of the bright cavity mode in order to maximize the coherent cavity-magnon coupling strength g_{am} . A weak microwave probe is injected into the cavity and the transmitted or reflected field is monitored using phase-sensitive microwave detection. The measured output spectrum contains the hybridized cavity-magnon resonances, whose frequency splitting provides the spectroscopic readout channel for the gravitationally induced magnon-frequency shift. An additional microwave pump is used to control the effective magnon damping rate, allowing the system to be tuned into a gain-assisted non-Hermitian regime, as discussed in Sec. V. The device parameters and the general cavity-magnon configuration are chosen to be compatible with experimentally demonstrated YIG microwave-cavity platforms.

IV. MAGNON RESPONSE TO THE GRAVITY GRADIENT

The gravitational interaction described in the previous section manifests as a perturbative shift in the magnon

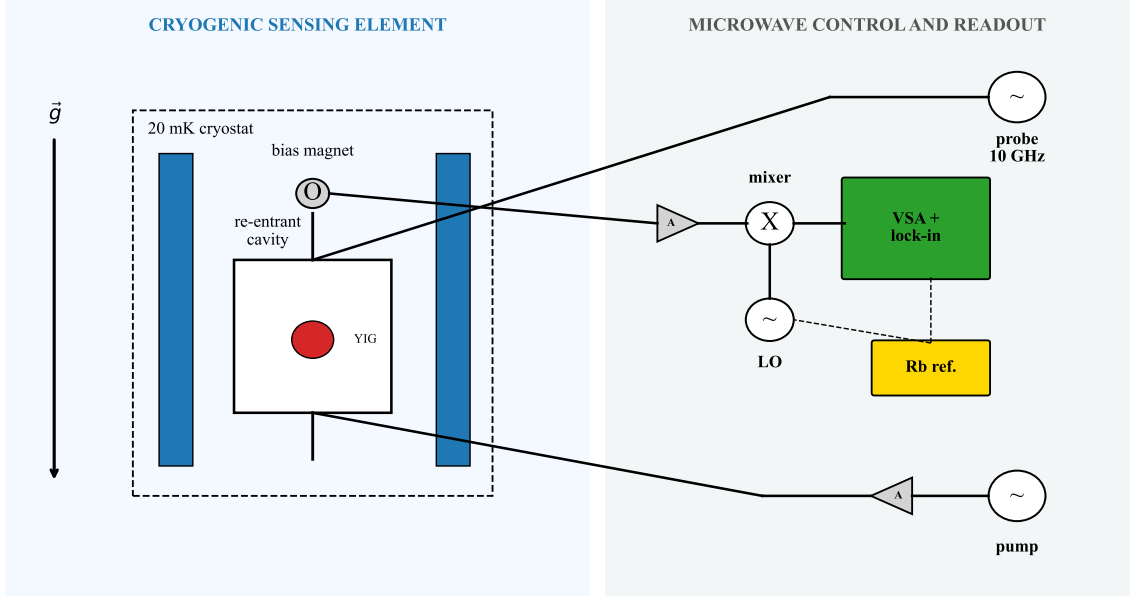


FIG. 1: Schematic of the gain-assisted cavity-magnon gravity-gradient sensor. A YIG crystal is placed inside a two-post re-entrant microwave cavity and biased by a static magnetic field to tune the uniform magnon mode close to the cavity resonance. A weak microwave probe interrogates the hybridized cavity-magnon spectrum, while an external microwave pump controls the effective magnon damping rate and enables operation in the gain-assisted non-Hermitian regime. The transmitted or reflected microwave field is measured using phase-sensitive detection referenced to a stable local oscillator.

resonance frequency. One of the most commonly used observables in quantum sensing is precisely such a frequency shift [26, 27]. In this section, we calculate the gravitationally induced frequency shift of the magnon mode and show how it can be used to extract the gravity-gradient parameter.

A. Frequency Shift of the Magnon

Variations in the gravitational field lead to changes in both the resonance frequency and the damping rate of the magnon mode. In the frequency domain, the linearized equation of motion for the magnon operator, derived from Eq. (26), can be written as [28]

$$-i\omega\hat{b} = -i\omega_m\hat{b} - ig_{\text{am}}\hat{a} - i\mu_B B_z \left[2 - \frac{gl_z^2}{5R} \right] \hat{b} - \frac{\kappa_m}{2}\hat{b} + \sqrt{\kappa_m}\hat{b}_{\text{in}}. \quad (27)$$

Here, $\hat{b}_{\text{in}}$ represents the noise fluctuations introduced by the environment and satisfies the standard correlation functions

$$\langle \hat{b}_{\text{in}}(t)\hat{b}_{\text{in}}^\dagger(t') \rangle = (n_{\text{th}}^m + 1)\delta(t - t'), \quad (28)$$

where n_{th}^m is the thermal occupation number of the magnon bath, $\delta(t - t')$ represents the delta-correlated

nature of the noise, assuming Markovian thermal fluctuations. In the other hand, equation (25) in the frequency frame are

$$\left[-i(\omega - \omega_a) + \frac{\kappa_a}{2} \right] \hat{a} + g_{\text{am}}\hat{b} = \sqrt{\kappa_a}\hat{a}_{\text{in}}, \quad (29)$$

where $\hat{a}_{\text{in}}$ represents the vacuum noise input for the cavity mode, satisfying the correlation functions [25, 29, 30]

$$\langle a_{\text{in}}^\dagger(\omega)a_{\text{in}}(\omega') \rangle = 2\pi n_{\text{th}}^{(a)} \delta(\omega - \omega'), \quad (30)$$

where $n_{\text{th}}^a = [e^{\hbar\omega_a/k_B T} - 1]^{-1}$ is the thermal occupation number of the photon bath, where k_B is the Boltzmann constant and T is the temperature. This term accounts for the fluctuations arising from the coupling between the cavity and its external bath. Finally, we can define the coefficient of the magnon operator as

$$[\chi_m^{-1} + \Sigma(\omega)] \hat{b} = \sqrt{\kappa_m}\hat{b}_{\text{in}}, \quad (31)$$

where the self-energy is the modification of the linear response to an gravitational change, that consist the susceptibility photon in microwave cavity, which can be expressed as [27, 28]

$$\Sigma(\omega) = -g_{\text{am}}^2 \chi_a + i\mu_B B_z \left(2 - \frac{gl_z^2}{5Rc^2} \right), \quad (32)$$

where the cavity susceptibility is

$$\chi_a = \frac{1}{-i(\omega - \omega_a) + \frac{\kappa_a}{2}}, \quad (33)$$

and susceptibility of the magnons in YIG which can be expressed [27]

$$\chi_m(\omega) = \frac{1}{-i(\omega - \omega_m) + \kappa_m/2}, \quad (34)$$

where consist the damping rate and frequency of magnons. The real and imaginary parts of the self-energy determine the effective magnon frequency and damping rate. The explicit expressions are $\Omega_m = \omega_m + \text{Im}\Sigma(\omega)$ is frequency shift and $\kappa_{\text{eff}} = \kappa_m + \text{Re}\Sigma(\omega)$ is effective damping rate, Therefore, the equation (35) becomes as follows

$$\left[-i(\omega - \Omega_m) + \frac{\kappa_{\text{eff}}}{2}\right] \hat{b} = \sqrt{\kappa_m} \hat{b}_{in}, \quad (35)$$

Following the standard optomechanical approach [28], the effective shifted magnon frequency Ω_m is given by

$$\Omega_m = \omega_m - \frac{g_{\text{am}}^2(\kappa_a/2)}{(\omega - \omega_a)^2 + \kappa_a^2/4} + \frac{\mu_B B_z}{\hbar} \left[2 - \frac{gl_z^2}{5Rc^2}\right], \quad (36)$$

For convenience, we isolate the contribution arising from the gravitational field and denote it by δ_0 , defined as

$$\delta_0 = \frac{\mu_B B_z}{\hbar} \left[2 - \frac{gl_z^2}{5Rc^2}\right]. \quad (37)$$

The quantity δ_0 represents the gravitationally induced shift of the magnon resonance frequency. Although this shift is much smaller than the bare magnon and cavity frequencies, it provides a direct perturbation channel through which the gravity-gradient signal can be encoded in the magnon mode. In the following, we denote this small frequency perturbation by δ_g when analyzing its effect on the coupled cavity-magnon spectrum. For a conventional linear sensor, such a weak frequency perturbation would produce only a correspondingly small shift in the measured resonance. To enhance the spectral response, we therefore consider a gain-assisted non-Hermitian cavity-magnon system operated close to an exceptional point.

V. THE GAIN-ASSISTED NON-HERMITIAN SYSTEM

Detecting extremely weak perturbations, such as gravitationally induced frequency shifts, requires a sensing mechanism with enhanced responsiveness beyond conventional linear regimes. In this context, non-Hermitian systems operating near EP have attracted significant attention due to their nontrivial spectral topology and enhanced response to external perturbations [31]. An exceptional point is a degeneracy in a non-Hermitian system where both eigenvalues and eigenvectors coalesce, leading to a non-analytic dependence of the eigenfrequencies on perturbations. In contrast to standard Hermitian

systems, where the response is typically linear, the eigenfrequency splitting near an EP follows a fractional power-law scaling with respect to the perturbation strength, which can significantly enhance the system response [31]. To exploit this enhanced sensitivity, we consider a gain-assisted hybrid magnon-cavity system engineered to operate at a second-order EP. In the following, we derive the effective non-Hermitian Hamiltonian of the system and establish the conditions required to access the EP regime.

A. Hamiltonian Formulation

We consider a hybrid cavity-magnon system consisting of a YIG ellipsoid coupled to a microwave cavity mode. Such systems are suitable platforms for studying non-Hermitian physics and exceptional points [32–34]. Starting from the Hamiltonian in Eq. (24) and the Langevin equations in Eqs. (25) and (26), we now write a simplified two-mode model for the cavity and magnon resonances. For finding the eigenfrequencies, we keep the coherent coupling and the damping terms, and we omit the input-noise operators. The noise terms will be included later when calculating the noise spectrum and the signal-to-noise ratio. In a frame rotating at the drive frequency ω_d , and after applying the rotating-wave approximation, the cavity-magnon coupling becomes $g_{\text{am}}(\hat{a}^\dagger \hat{b} + \hat{a} \hat{b}^\dagger)$. This term describes the exchange of excitations between the cavity photon mode and the magnon mode. The cavity loss is described by the decay rate κ_a , while the magnon mode is described by an effective linewidth κ_{eff} . These damping terms are included as imaginary parts of the mode frequencies. Therefore, the effective non-Hermitian Hamiltonian is

$$\hat{H} = \Delta_a \hat{a}^\dagger \hat{a} + \Delta_m \hat{b}^\dagger \hat{b} + g_{\text{am}}(\hat{a}^\dagger \hat{b} + \hat{a} \hat{b}^\dagger) - i \frac{\kappa_a}{2} \hat{a}^\dagger \hat{a} - i \frac{\kappa_{\text{eff}}}{2} \hat{b}^\dagger \hat{b}, \quad (38)$$

where $\Delta_a = \omega_a - \omega_d$ and $\Delta_m = \omega_m - \omega_d$ are the cavity and magnon detunings from the drive frequency, respectively. The parameter κ_a is the cavity decay rate, while κ_{eff} is the effective linewidth of the magnon mode. Without gain, $\kappa_{\text{eff}} = \kappa_m$, where κ_m is the intrinsic magnon damping rate. In the gain-assisted regime, the magnon damping is partially compensated by a microwave pump. This can be described as

$$\kappa_{\text{eff}} = \kappa_m - G, \quad (39)$$

where G is the pump-induced magnon gain rate. Physically, such gain can be realized through parametric pumping of the magnetic medium, which amplifies the magnon population and acts as an effective negative damping of the magnon mode [35, 36]. Thus, increasing G reduces the net magnon linewidth, and when $G > \kappa_m$, the magnon mode becomes effectively amplifying. The corresponding equations for the two mode amplitudes can

be written as

$$\frac{d}{dt} \begin{pmatrix} \hat{a} \\ \hat{b} \end{pmatrix} = -i \begin{pmatrix} \Delta_a - i\kappa_a/2 & g_{am} \\ g_{am} & \Delta_m - i\kappa_{\text{eff}}/2 \end{pmatrix} \begin{pmatrix} \hat{a} \\ \hat{b} \end{pmatrix}. \quad (40)$$

Accordingly, the matrix appearing in Eq. (40) defines the effective non-Hermitian Hamiltonian of the system in the (a, b) basis,

$$H_{\text{eff}} = \begin{pmatrix} \Delta_a - i\frac{\kappa_a}{2} & g_{am} \\ g_{am} & \Delta_m - i\frac{\kappa_{\text{eff}}}{2} \end{pmatrix}. \quad (41)$$

The diagonal elements describe the detunings and effective linewidths of the uncoupled cavity and magnon modes, while the off-diagonal elements describe the coherent cavity-magnon coupling. We use this two-mode form to find the exceptional-point condition and then study how a small gravitationally induced magnon-frequency shift changes the hybrid-mode spectrum.

B. Condition for the Exceptional Point

We now use the effective Hamiltonian in Eq. (38) to determine the exceptional-point condition. As discussed above, the microwave pump controls the effective magnon linewidth through $\kappa_{\text{eff}} = \kappa_m - G$. By increasing the pump-induced gain G , the magnon loss can be reduced and eventually converted into an effective gain. The balanced gain-loss condition is reached when

$$\kappa_{\text{eff}} = -\kappa_a. \quad (42)$$

At resonance, $\Delta_a = \Delta_m = 0$, this condition reduces the effective Hamiltonian matrix to

$$H_{\text{eff}} = \begin{pmatrix} -i\frac{\kappa_a}{2} & g_{am} \\ g_{am} & i\frac{\kappa_a}{2} \end{pmatrix}. \quad (43)$$

This matrix has the standard balanced gain-loss form of a two-mode non-Hermitian system. In this ideal limit, it can be interpreted as a $\mathcal{PT}$ -symmetric effective Hamiltonian, where the cavity mode provides loss and the magnon mode provides effective gain [32, 33].

The complex eigenfrequencies are obtained from the characteristic equation of Eq. (43), giving

$$\lambda_{\pm} = \pm \sqrt{g_{am}^2 - \left(\frac{\kappa_a}{2}\right)^2}. \quad (44)$$

The system behavior is therefore determined by the competition between the coherent coupling strength g_{am} and the gain-loss scale $\kappa_a/2$. When $g_{am} > \kappa_a/2$, the square root in Eq. (44) is real and the two hybrid modes remain spectrally split. This corresponds to the unbroken $\mathcal{PT}$ -symmetric regime, where balanced gain-loss systems can exhibit real eigenfrequencies [32, 33]. When $g_{am} < \kappa_a/2$, the square root becomes imaginary and the system enters the broken $\mathcal{PT}$ -symmetric regime, where the two modes acquire different imaginary eigenfrequency components

associated with effective amplification and damping [32–34]. The boundary between these two regimes occurs when the radicand in Eq. (44) vanishes,

$$g_{am} = \frac{\kappa_a}{2}. \quad (45)$$

At this point, the two eigenvalues coalesce at $\lambda_+ = \lambda_- = 0$. The corresponding eigenvectors also merge, so the degeneracy is an exceptional point rather than an ordinary Hermitian degeneracy. Near this point, a small perturbation to the magnon frequency produces a non-analytic change in the hybrid eigenfrequencies. This square-root response is the mechanism used below to amplify the spectral effect of the gravitationally induced magnon-frequency shift.

VI. GRAVITATIONAL PERTURBATION AMPLIFICATION AT THE EP

Building on the EP condition defined in Eq. (45), we now evaluate how a gravitationally induced magnon-frequency shift modifies the hybrid-mode spectrum.

A. Perturbed Hamiltonian

The gravitational gradient induces a small shift of the magnon resonance frequency. In the effective two-mode basis $(\hat{a}, \hat{b})$, the first component represents the cavity photon mode and the second component represents the magnon mode. Since the gravitational interaction derived in Sec. II modifies the magnon frequency, its leading effect is represented as a perturbation to the magnon diagonal element of the effective Hamiltonian. We do not include additional perturbations to the cavity resonance or to the cavity magnon coupling, because the gravitational term is modeled here as a frequency shift of the magnon mode only. At the EP, where $g_{am} = \kappa_a/2$, the perturbed effective Hamiltonian is therefore written as

$$H_{\text{pert}} = H_{\text{eff}} + \begin{pmatrix} 0 & 0 \\ 0 & \delta_g \end{pmatrix} = \begin{pmatrix} -i\frac{\kappa_a}{2} & \frac{\kappa_a}{2} \\ \frac{\kappa_a}{2} & i\frac{\kappa_a}{2} + \delta_g \end{pmatrix}. \quad (46)$$

Here, δ_g denotes the gravitationally induced magnon-frequency shift introduced in Eq. (37). The zeros in the perturbation matrix indicate that, within this approximation, gravity does not directly shift the cavity resonance and does not modify the coherent coupling g_{am} . Its effect is kept only as a small correction to the magnon resonance frequency. To determine the modified eigenfrequencies, we solve the characteristic equation $\det(H_{\text{pert}} - \lambda \mathbf{I}) = 0$. Using Eq. (46), we obtain

$$\det \begin{pmatrix} -i\kappa_a/2 - \lambda & \kappa_a/2 \\ \kappa_a/2 & i\kappa_a/2 + \delta_g - \lambda \end{pmatrix} = 0. \quad (47)$$

Expanding the determinant gives

$$\left(-i\frac{\kappa_a}{2} - \lambda\right) \left(i\frac{\kappa_a}{2} + \delta_g - \lambda\right) - \left(\frac{\kappa_a}{2}\right)^2 = 0. \quad (48)$$

After simplification, this becomes

$$\lambda^2 - \delta_g \lambda - i \frac{\kappa_a}{2} \delta_g = 0. \quad (49)$$

The exact solutions are therefore

$$\lambda_{\pm} = \frac{\delta_g}{2} \pm \frac{1}{2} \sqrt{\delta_g^2 + 2i\kappa_a \delta_g}. \quad (50)$$

For the gravitational perturbations considered here, $|\delta_g| \ll \kappa_a$. In this limit, the term δ_g^2 inside the square root is negligible compared with $2i\kappa_a \delta_g$. The common shift $\delta_g/2$ is also much smaller than the square-root splitting. Thus, to leading order,

$$\lambda_{\pm} \approx \pm \sqrt{i \frac{\kappa_a}{2} \delta_g}. \quad (51)$$

Equation (51) is the characteristic square-root response of a second-order exceptional point. In an ordinary linear sensor, a small perturbation produces a frequency shift proportional to δ_g . At the EP, however, the two coalesced eigenfrequencies separate as $\Delta\lambda \propto \sqrt{\delta_g}$, which is the standard non-analytic eigenvalue response of an EP [32–34]. To make this response explicit, we define the EP-induced mode splitting as the separation between the two perturbed eigenfrequencies,

$$\Delta\lambda \equiv |\lambda_+ - \lambda_-|. \quad (52)$$

Using Eq. (44), this splitting is

$$\Delta\lambda \approx 2 \left| \sqrt{i \frac{\kappa_a}{2} \delta_g} \right| = \sqrt{2\kappa_a |\delta_g|}. \quad (53)$$

Thus, the gravitationally induced magnon-frequency shift is converted into a larger hybrid-mode splitting near the EP. This provides a spectral amplification mechanism, which we quantify by comparing the EP-induced splitting with the original perturbation:

$$\eta = \left| \frac{\Delta\lambda}{\delta_g} \right| \approx \sqrt{\frac{2\kappa_a}{|\delta_g|}}. \quad (54)$$

Here, η measures how much larger the EP-induced spectral splitting is compared with a conventional linear frequency shift of the same magnitude. It should not be interpreted as the final measurement sensitivity, since the actual detectability also depends on the noise floor and the signal-to-noise ratio. Using the parameters in Table ??, with $\kappa_a = 9.4 \times 10^6 \text{ s}^{-1}$ and $|\delta_g| = 6 \times 10^{-17} \text{ s}^{-1}$, the expected separation between the two hybrid modes is

$$\frac{\Delta\lambda}{2\pi} \approx \frac{1}{2\pi} \sqrt{2(9.4 \times 10^6)(6 \times 10^{-17})} \approx 5 \times 10^{-6} \text{ Hz}. \quad (55)$$

This shows that a gravitationally induced magnon-frequency shift at the 10^{-17} s^{-1} scale can appear in the hybrid spectrum as a microhertz-scale separation between the two EP-split modes.

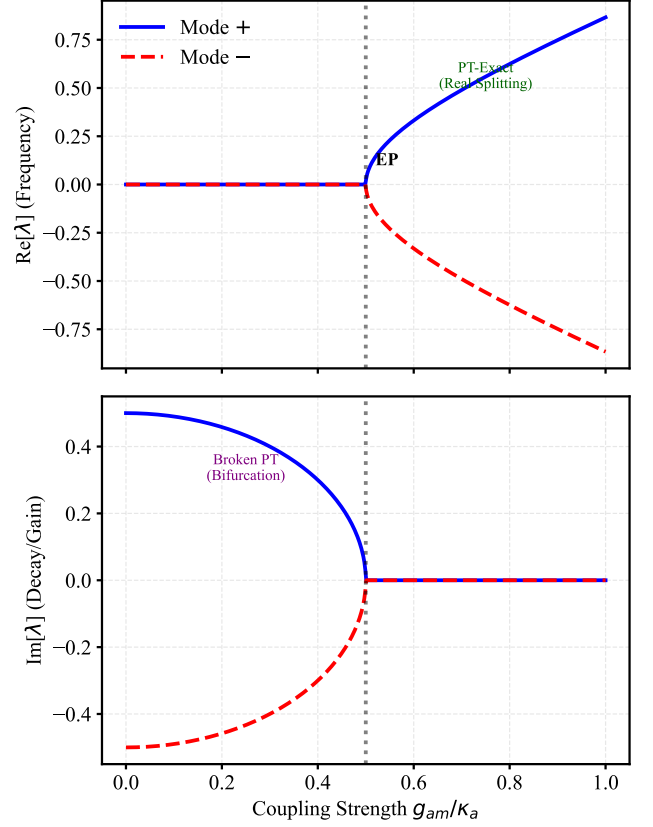


FIG. 2: Eigenvalue landscape of the gain-assisted magnonic system. (Top) The real part of the eigenfrequencies $\text{Re}[\lambda]$ as a function of the normalized coupling rate g_{am}/κ_a . The coalescence of the modes at $g_{am} = 0.5\kappa_a$ marks the exceptional point. (Bottom) The imaginary part $\text{Im}[\lambda]$, illustrating the bifurcation into gain and loss modes within the broken $\mathcal{PT}$ -symmetric phase. The sensor is operated exactly at the EP singularity to maximize its susceptibility to external perturbations.

VII. QUANTUM NOISE

Although operation near an exceptional point enhances the spectral response to a small gravitational perturbation, it also changes the noise properties of the coupled cavity-magnon modes. A complete evaluation of the sensor must therefore include both the enhanced mode splitting and the uncertainty with which this splitting can be measured.

The relevant noise sources can be divided into two groups. The first group consists of fundamental fluctuations, including cavity input noise, magnon thermal noise, photon shot noise, and the excess noise associated with the non-orthogonality of the eigenmodes near the exceptional point. The second group consists of technical fluctuations, such as electronic $1/f$ noise, slow thermal drift, magnetic-field fluctuations, and mechanical instability.

A. Petermann Factor

Near an exceptional point, the right and left eigenvectors of a non-Hermitian system become strongly non-orthogonal. This non-orthogonality increases the noise associated with the hybrid modes and is commonly quantified by the Petermann factor [32, 33, 37],

$$K = \frac{\langle \psi_L | \psi_L \rangle \langle \psi_R | \psi_R \rangle}{|\langle \psi_L | \psi_R \rangle|^2}, \quad (56)$$

where $|\psi_R\rangle$ and $\langle \psi_L|$ are the right and left eigenvectors of the effective non-Hermitian Hamiltonian, respectively.

For orthogonal modes, $K = 1$. As the system approaches the exceptional point, the overlap $\langle \psi_L | \psi_R \rangle$ decreases and K increases. The Petermann factor therefore represents an excess-noise penalty associated with operating close to the eigenvector coalescence.

For the perturbed two-mode Hamiltonian considered here, the Petermann factor may be expressed in terms of the perturbed eigenvectors, or equivalently evaluated numerically from Eq. (56). In the following sensitivity estimate, its effect is included through the factor $\sqrt{K}$, because K describes the enhancement of the noise power, whereas the corresponding noise amplitude scales as $\sqrt{K}$.

B. Frequency Uncertainty

The gravitational signal is measured through the separation of the two hybrid-mode resonances. The precision with which the resonance frequencies can be determined is limited by the effective linewidth and by the number of detected photons.

For a measurement containing N_{tot} detected photons, the quantum-limited frequency uncertainty can be estimated as

$$\sigma_f \approx \frac{\kappa_a}{2\pi\sqrt{N_{\text{tot}}}}\sqrt{K}, \quad (57)$$

where $\kappa_a/2\pi$ is the cavity linewidth in hertz. The factor $1/\sqrt{N_{\text{tot}}}$ represents the reduction of photon shot noise with increasing measurement time and input photon flux, while $\sqrt{K}$ accounts for the excess noise near the exceptional point.

Equation (57) shows that the exceptional point does not provide a noise-free enhancement. The spectral splitting increases near the EP, but the frequency uncertainty also increases because of the Petermann factor. The net performance must therefore be evaluated through the signal-to-noise ratio rather than through the spectral amplification factor alone.

C. Suppression of Low-Frequency Technical Noise

In addition to the quantum noise considered above, a nearly static gravitational signal can be masked by low-frequency technical noise. Important examples include

electronic $1/f$ noise, slow thermal drift, magnetic-field instability, and long-term fluctuations of the cavity resonance.

As an optional noise-reduction method, the sensing part of the setup, consisting of the cavity, the YIG sample, and the bias magnet, can be rotated slowly with respect to the direction of the Earth's gravitational field. The microwave control and readout electronics may remain stationary. The rotation periodically changes the projection of the gravitational field onto the sensitive axis of the YIG sample, so that the gravitational variation can be written as

$$\delta g(t) = \delta g \cos(\Omega t), \quad (58)$$

where δg is the amplitude of the gravitational variation and Ω is the angular modulation frequency.

This modulation transfers the signal from zero frequency to a finite frequency Ω , where the contribution of $1/f$ noise and slow drift is generally smaller. The modulated response can then be recovered using phase-sensitive or lock-in detection, following the same general principle used in precision modulation measurements and rotating gravity gradiometers [38, 39]. The rotation is not required for the basic exceptional-point sensing mechanism and is introduced only as a possible method for reducing low-frequency technical noise.

VIII. SENSITIVITY

We now compare the gravitationally induced spectral splitting with the frequency uncertainty derived in Sec. VII. The spectral amplification factor introduced previously describes the response of the eigenfrequencies, but the practical performance of the sensor is determined by the signal-to-noise ratio and by the smallest gravitational variation that can be resolved.

A. Photon Flux

The number of photons detected during an integration time τ is estimated as

$$N_{\text{tot}} = \frac{P_{\text{in}}\tau}{\hbar\omega_a}, \quad (59)$$

where P_{in} is the microwave probe power and ω_a is the cavity resonance frequency.

For $P_{\text{in}} = 20$ mW, $\tau = 1000$ s, and $\omega_a = 2\pi \times 10$ GHz, this gives

$$N_{\text{tot}} \approx \frac{(20 \times 10^{-3})(1000)}{(1.05 \times 10^{-34})(2\pi \times 10^{10})} \approx 3 \times 10^{24}. \quad (60)$$

The large photon number reduces the shot-noise contribution to the uncertainty in the fitted resonance frequencies. This estimate assumes that the probe remains within the linear-response regime and does not significantly heat the cavity or the YIG sample.

B. Signal-to-Noise Ratio

The measurable signal is the separation between the two perturbed hybrid-mode resonances,

$$\Delta f = \frac{\Delta \lambda}{2\pi}. \quad (61)$$

The signal-to-noise ratio is therefore defined as

$$\text{SNR} = \frac{\Delta f}{\sigma_f}, \quad (62)$$

where σ_f is the frequency uncertainty given in Eq. (57).

Using the present parameter set, the EP-induced mode splitting is of the order of a few microhertz, while the estimated frequency uncertainty remains much larger. Taking

$$\Delta f \approx 5 \times 10^{-6} \text{ Hz} \quad (63)$$

and the current estimate

$$\sigma_f \approx 0.32 \text{ Hz}, \quad (64)$$

we obtain

$$\text{SNR} \approx \frac{5 \times 10^{-6}}{0.32} \approx 1.6 \times 10^{-5}. \quad (65)$$

The present conservative parameter set therefore does not yet provide a directly resolvable signal. The SNR can be improved by reducing the cavity linewidth, increasing the detected photon number, increasing the gravitational coupling through the YIG geometry, and reducing technical noise through stabilization or optional modulation.

A lower cavity linewidth directly reduces the frequency uncertainty in Eq. (57). Increasing the YIG dimensions or geometric anisotropy increases the factors l_x^2 , l_y^2 , and l_z^2 , and therefore strengthens the gravity-induced magnon-frequency shift. Longer integration times also increase N_{tot} , provided that the system remains stable over the full measurement interval.

An optimized implementation should aim for

$$\text{SNR} \gtrsim 10, \quad (66)$$

so that the gravitationally induced splitting is clearly distinguishable from the measurement uncertainty rather than merely reaching the threshold $\text{SNR} = 1$.

C. Relative Gravity Resolution

The smallest resolvable gravitational variation is denoted by δg_{min} . It is determined by the frequency uncertainty and by the response of the measured splitting to gravity:

$$\delta g_{\text{min}} = \frac{\sigma_f}{|\partial \Delta f / \partial g|}. \quad (67)$$

A convenient dimensionless measure of the sensor precision is the fractional gravity resolution

$$\frac{\delta g_{\text{min}}}{g}, \quad (68)$$

where g is the local gravitational acceleration. A smaller value of $\delta g_{\text{min}}/g$ corresponds to a more precise measurement of gravity.

For the second-order EP response, the measured splitting scales as

$$\Delta f \propto \sqrt{g}. \quad (69)$$

Therefore,

$$\frac{\partial \Delta f}{\partial g} = \frac{\Delta f}{2g}. \quad (70)$$

Substituting this relation into Eq. (67) gives

$$\frac{\delta g_{\text{min}}}{g} = \frac{2\sigma_f}{\Delta f} = \frac{2}{\text{SNR}}. \quad (71)$$

Equation (71) directly connects the gravity resolution to the measurable signal-to-noise ratio. For the current conservative value $\text{SNR} \approx 1.6 \times 10^{-5}$, the fractional uncertainty is much larger than unity, confirming that the present parameters are insufficient for a gravity measurement. In an optimized configuration with $\text{SNR} \approx 10$, the corresponding fractional resolution would be

$$\frac{\delta g_{\text{min}}}{g} \approx 0.2. \quad (72)$$

Further improvement requires a larger SNR through a narrower cavity linewidth, stronger magnon-gravity coupling, greater detected photon flux, and improved rejection of low-frequency technical noise.

Appendix A: Geometric Parameters of the YIG Ellipsoid

The squared geometric parameters l_x^2 , l_y^2 , and l_z^2 appearing in the magnon-gravity Hamiltonian are defined in terms of the YIG ellipsoid principal semi-axes (a , b , and c) as follows

$$l_x^2 \equiv ac(a^2 - c^2) + b^2(a^2 - c^2) + ab(a^2 - b^2) + ac(a^2 - c^2) - b^2(a^2 - c^2), \quad (A1)$$

$$l_y^2 \equiv ac(a^2 - c^2) + bc(b^2 - c^2) + ab(a^2 - b^2) + b^2(a^2 - c^2) + bc(b^2 - c^2) - a^2(7 + 9a^2 - b^2 - 2c^2) - b^2(7 - a^2 + 9b^2 - 2c^2), \quad (A2)$$

$$l_z^2 \equiv a^2(7 + 9a^2 - b^2 - 2c^2) + 2ab(a^2 - b^2) + b^2(7 - a^2 + 9b^2 - 2c^2) - a^2(-14 + 9a^2 + b^2 - 4c^2) - b^2(-14 + a^2 + 9b^2 - 4c^2). \quad (A3)$$

As evident from these definitions, the disparities among l_x , l_y , and l_z stem directly from the broken spherical symmetry of the triaxial YIG ellipsoid ($a \neq b \neq c$). This geometric anisotropy dictates the spatial dependency of the system, translating mathematically into distinct, direction-dependent coupling strengths for the magnon-gravity interaction along the three principal axes.

Appendix B: The second-order exceptional point

In this appendix, we provide a minimal example of a second-order exceptional point (EP2) in order to clarify the physical mechanism used in the main text. An exceptional point is a degeneracy of a non-Hermitian system at which not only the eigenvalues, but also the corresponding eigenvectors, coalesce. This is different from an ordinary Hermitian degeneracy, where two eigenvalues may become equal while the eigenvectors remain linearly independent.

Consider two coupled modes with balanced loss and gain. After removing a common resonance frequency, the effective non-Hermitian Hamiltonian can be written as

$$H_{\text{EP}} = \begin{pmatrix} -i\gamma/2 & g \\ g & i\gamma/2 \end{pmatrix}, \quad (\text{B1})$$

where g is the coherent coupling rate between the two modes and γ characterizes the loss-gain contrast. The eigenvalues are obtained from

$$\det(H_{\text{EP}} - \lambda I) = 0, \quad (\text{B2})$$

which gives

$$\lambda_{\pm} = \pm \sqrt{g^2 - \left(\frac{\gamma}{2}\right)^2}. \quad (\text{B3})$$

For $g > \gamma/2$, the eigenvalues are real and the two modes form two distinct hybridized normal modes. For $g < \gamma/2$, the eigenvalues become imaginary, corresponding to one amplifying and one decaying mode. The boundary between these two regimes occurs at

$$g = \frac{\gamma}{2}. \quad (\text{B4})$$

At this point, Eq. (B3) gives

$$\lambda_+ = \lambda_- = 0. \quad (\text{B5})$$

However, this is not merely an eigenvalue degeneracy. Substituting $g = \gamma/2$ into Eq. (B1), the eigenvector equation $H_{\text{EP}}|\psi\rangle = 0$ gives

$$|\psi_{\text{EP}}\rangle \propto \begin{pmatrix} 1 \\ i \end{pmatrix}. \quad (\text{B6})$$

Both eigenvectors collapse into this single vector. Therefore, the Hamiltonian becomes defective and cannot be diagonalized in the usual way. This simultaneous coalescence of eigenvalues and eigenvectors is the defining property of an exceptional point.

The relevance of the EP to sensing becomes clear when a small perturbation is added to one of the two modes. For example, let a weak frequency perturbation ϵ shift the second mode:

$$H_{\text{pert}} = \begin{pmatrix} -i\gamma/2 & \gamma/2 \\ \gamma/2 & i\gamma/2 + \epsilon \end{pmatrix}. \quad (\text{B7})$$

The characteristic equation becomes

$$\lambda^2 - \epsilon\lambda - i\frac{\gamma}{2}\epsilon = 0. \quad (\text{B8})$$

For a small perturbation $|\epsilon| \ll \gamma$, the leading-order eigenvalues are

$$\lambda_{\pm} \simeq \pm \sqrt{i\frac{\gamma}{2}\epsilon}. \quad (\text{B9})$$

Thus, the eigenfrequency splitting scales as

$$\Delta\lambda \propto \sqrt{\epsilon}. \quad (\text{B10})$$

This square-root response is the key feature of EP-assisted sensing. In an ordinary sensor, a small perturbation normally produces a linear shift, $\Delta\lambda \propto \epsilon$. At an EP2, the same perturbation produces a much larger relative response for $|\epsilon| \ll 1$, because $\sqrt{\epsilon} \gg \epsilon$.

In the main text, the two modes correspond to the microwave cavity mode and the magnon mode. The cavity provides the lossy mode, while the effective magnon linewidth can be tuned by the gain pump. When the gain-assisted magnon mode balances the cavity loss, the system is tuned close to the EP condition. The gravitational gradient then acts as a small perturbation to the magnon frequency, playing the role of ϵ in Eq. (B7). Therefore, the gravitationally induced magnon-frequency shift is converted into an enhanced square-root eigenfrequency splitting, which is the central sensing mechanism used in this work.

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